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3 November 1979
Superseding
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3 November 1978

MILITARY SPECIFICATION

RECEIVERS, RANGE SAFETY, COMMAND

This specification is approved for use by the US Army Test and Evaluation Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification sets forth the requirements for range safety command receivers, to be used in conjunction with ground based frequency modulated radio transmitters, in missile flight termination systems. The receivers will be used with a variety of missiles and are required to function reliably under conditions of extreme temperatures, shock and vibration.

1.2 Classification. The receiver, range safety command, shall be of the following types and classes:

Type I. - Designates units having a volume of 30 cubic inches or more.
Type II. - Designates units having a volume of less than 30 cubic inches.
Class 1. - Designates units wherein the output section incorporates relay operated outputs.
Class 2. - Designates units incorporating semiconductor type outputs such as transistors and silicon controlled rectifiers.
* Class 3. - Designates units incorporating both relay and semiconductor type outputs.

1.2.1 Style. Style designation will be assigned to units of the same type and class to denote differences in construction, design, or appearance. Specific construction differences such as connector types required will be identified by style designation. Style designation will be of the form: Single capital Gothic letter followed by an Arabic numeral. Example: Style A1.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Test and Evaluation Command, ATTN: DRSTE-AD-M, Aberdeen Proving Ground, MD 21005 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1430

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2. APPLICABLE DOCUMENTS

2.1 Issues of document. The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-R-87954/1	- Receiver, Range Safety, Command Type I, Class 1, Style A1

STANDARDS

FEDERAL

FED-STD-356	- Commercial Packaging of Supplies and Equipment

MILITARY

MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurements of
MIL-STD-810	- Environmental Test Methods

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Specification sheets. The individual requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of conflict between requirements of this specification and the specification sheet, the latter shall govern.

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3.2 Classification of requirements. The requirements for the product are classified herein as follows:

Requirement	Paragraph
Qualification	3.3
First Article	3.4
Design	3.5
Reliability	3.6
Construction	3.7
Performance Characteristic	3.8
Environmental	3.9
Electromagnetic Interference Suppression	3.10
Physical Characteristics	3.11
Identification	3.12
Workmanship	3.13
Technical Data	3.14

3.3 Qualifications. Receivers furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids.

3.4 First article. When specified (see 6.4), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.).

3.5 Design. The receiver shall be functionally configured in three sections: the radio frequency (RF)/discriminator section; the decoder section; and the output section (see figure 1).

3.5.1 RF/Discriminator section.

3.5.1.1 Functional. This section shall consist of the necessary circuitry to process frequency modulated (FM)/radio frequency (RF) input signals in the ultra high frequency band (UHF) specified herein and provide detected FM to the decoder section of the unit. The modulation shall consist of discrete tone commands (see table 1) with nominal frequency deviations of $\pm$30 kHz per tone.

3.5.1.2 Operating frequency range. The receiver shall be tunable to any frequency between 406 mHz and 550 mHz, without replacement of any component except the local oscillator crystal(s) and the operating frequency preselector tuning elements. Unless otherwise specified, the receiver shall be furnished with crystal(s) and preselector tuning elements installed and set to the operating frequency specified in the ordering data.

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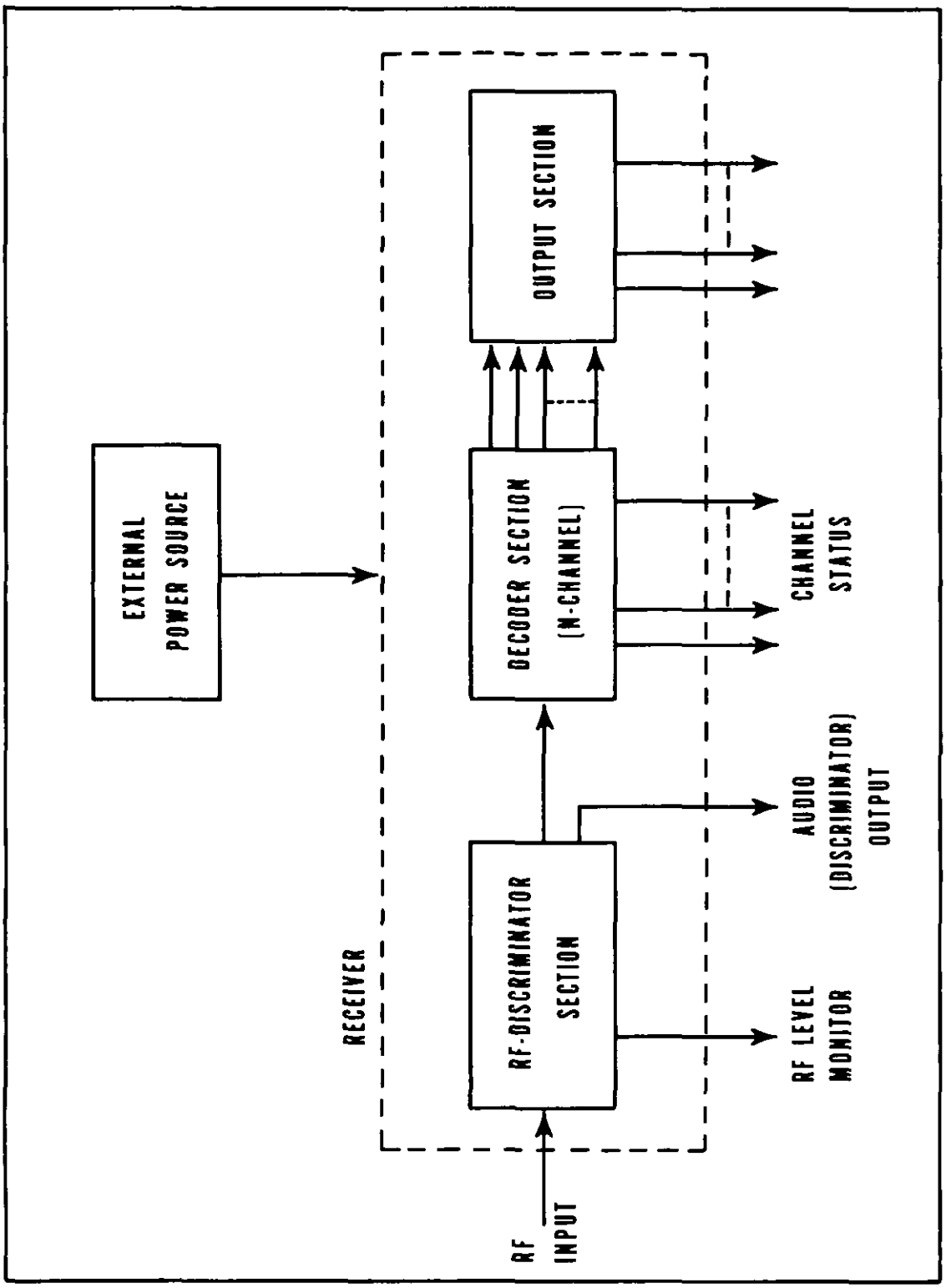


FIGURE 1. RANGE SAFETY COMMAND RECEIVER FUNCTIONAL CONFIGURATION.

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3.5.1.3 RF input level monitor. The design shall provide an RF input level monitor with output to provide a positive DC voltage not to exceed 5 volts; output proportional to the RF signal input and have the characteristics specified in the applicable specification sheet.

3.5.1.4 Audio output. The design shall provide an audio output of the detected FM when required by the specification sheets. The output characteristics shall be specified in the applicable specification sheets.

3.5.2 Command decoder section. The design shall provide a command decoder section which shall process a number of command channel frequencies to control the output section. The exact number of channels shall be set forth in the applicable specification sheet.

3.5.2.1 Function. The decoder section shall perform a "channel activated" or "channel deactivated" decision-making function for each of the required channels.

3.5.2.2 Command channel frequencies. The command channel frequencies shall be selected from Table I, in accordance with the instructions set forth in the ordering data (see 6.2). The number of channels shall be as set forth in the applicable specification sheet(s).

3.5.2.3 Command channel status monitor output. The design shall provide output(s), when specified by the specification sheet, to indicate channel status, that is "activated" or "deactivated." The output characteristics shall be in accordance with the specification sheet.

3.5.3 Output section. The design of the output section shall contain output circuitry controlled by command channel activation as detected in the command decoder section. The output section shall function as an integral part of the receiver. The output section shall have the characteristics set forth in the applicable specification sheet(s).

3.5.4 Power control circuit. When required by the applicable specification sheet, the design shall provide power control circuitry to remotely enable or disable the unit by using wire control circuits having the characteristics specified in the applicable specification sheet.

3.5.5 Single Point Failure. The design shall be such that any single failure shall not activate more than one command channel output. *

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TABLE I. Command channel frequency assignment.

Channel No.	Frequency (kHz)	Channel No.	Frequency (kHz)
1	7.50	11	25.01
2	8.46	12	28.21
3	9.54	13	31.83
4	10.76	14	35.90
5	12.14	15	40.49
6	13.70	16	45.68
7	15.45	17	51.52
8	17.43	18	58.12
9	19.66	19	65.56
10	22.17	20	73.95

3.6 Reliability. The mean number of operation cycles between failure of the receiver shall be at least 25,000, as demonstrated by passing the sequential reliability test described in Section 4. The contractor shall perform all maintenance and repair and furnish all replacement parts required throughout the reliability testing.

3.7 Construction.

3.7.1 Case finish and receiver construction. The case finish and receiver construction shall provide protection against corrosion, cracking, peeling and degradation of electrical connections and impedance paths necessary to insure performance within the requirements of this specification under all environmental conditions specified herein.

3.7.2 Connectors. The construction shall provide electrical connectors mounted and connected as specified in the applicable specification sheet.

3.8 Performance characteristics.

3.8.1 Predetection bandwidth. The overall predetection bandwith at the 3 dB and 60 dB points, and the reference receiver input level at which the bandwidth shall be measured is set forth in the applicable specification sheet. Passband shall be centered on the specified operating frequency within 20% of the measured bandwidth value at the above cited point.

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3.8.2 Threshold sensitivity. Required output section function shall be accomplished with an input signal at the assigned frequency ±0.005%, frequency modulated at the frequencies of the specified command channels (see 3.5.2.2) at ±30 kHz (±10%) peak deviation per tone and at the receiver RF input level specified in the applicable specification sheet.

3.8.3 Quieting sensitivity. With a receiver RF input level of -92 dbm (or as specified in applicable specification sheet) at the specified operating frequency, there shall be at least 20 db of quieting. (Quieting is defined as the ratio of the output voltage of the audio output, in the absence of RF signal, to the output voltage produced by the unmodulated signal.) *

3.8.4 Operating frequency tolerance. The receiver performance shall be within the limits specified for all operating carrier frequencies within ±0.005% of the value or values set forth in the ordering data (see 6.2).

3.8.5 RF signal level. The receiver shall operate within performance requirements throughout the RF input signal level range as specified in the applicable specification sheet.

3.8.6 Antenna impedance. The receiver shall operate within performance requirements when coupled to an antenna system having a voltage standing wave ratio (VSWR) not exceeding 2.0, referenced to 50 ohms. Termination of the receiver RF input in any impedance value shall not cause the receiver to become unstable, oscillate or be damaged.

3.8.7 Decoder frequency characteristics. With an RF input level of -92 dbm, the receiver shall have the following decoder performance, unless otherwise specified in the applicable specification:

a. Activate. The decoder shall make a "channel activated" decision, and the output section shall function as required by the applicable specification sheet, when the modulation level is within the range from ±15 kHz (±10%) to ±30 kHz (±10%) per channel peak deviation and the modulation frequency is within ±0.5% of the specified command channel frequency.

b. Deactivate. The decoder shall make a "channel deactivated" decision. The output section shall function as required by the applicable specification sheet when the modulation frequency is outside ±3.5% of the specified command channel frequency with a modulation level of ±40 kHz (±10%) peak deviation or less.

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3.8.8 Decoder signal plus noise characteristics. The noise response characteristics and "channel activated" decision threshold adjustment on each decoder channel shall provide the decoder performance specified below with the given signal plus noise composite modulation on the receiver input signal.

3.8.8.1 Decoder performance. When the "channel activated" decision is made no more than 50% of the time:

a. The command channel frequency level one db below the "channel activated" decision level with channel frequency is only the applied modulation.

b. The noise (uniform spectral density over the bandwidth required to pass all decoder channels) at an RMS level 12 db is greater than a. above.

3.8.9 Multiple command channels. Each of the number of command channels specified in the applicable specification sheets shall perform and initiate the output function(s) required by the specification sheet.

3.8.10 Response time. The minimum time between application of commands to the RF input signal (channel activation), at a specified level, and the corresponding output function shall be specified in the applicable specification sheet.

3.8.11 Power requirement.

3.8.11.1 Prime power. The receiver shall operate with a power source having the characteristics described in the applicable specification sheet.

3.8.11.2 Power consumption. The receiver power consumption shall be within the limits specified in the applicable specification sheet.

3.8.11.3 Reverse voltage protection. Receivers operating on direct currents shall not be damaged by application of voltages from 0 to 100 VDC reversed polarity at the prime power input.

3.8.11.4 Power controlled by output section. The power controlled by the output section to provide the specified output functions shall be set forth in the applicable specification sheet.

3.9 Environmental. The receiver shall operate within performance requirements under all possible combinations of the following conditions:

3.9.1 Acceleration. When subjected to constant acceleration of 1/ applied along the three major axes, both directions, for periods of five minutes along each axis in accordance with the Procedure II, Method 513.1, MIL-STD-810.

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3.9.2 Temperature-Altitude. Use Procedure I, Method 504, MIL-STD-810 under the following specified conditions:

a. Temperature: 1/

b. Altitude: 1/

c. Operating time: 1/

3.9.3 Shock. Use Procedure IV, Method 516, MIL-STD-810. The terminal peak sawtooth shock pulse configuration and its tolerance limits shall be as shown in Figure 516.2-1 of MIL-STD-810.

3.9.4 Vibration. The vibration environment shall be in accordance with Procedure V, Method 514, Table 514.2-V and Figure 514.2-5.

a. Sinusoidal curve: 1/

b. Random curve: 1/

3.9.5 Temperature. With the relative humidity ranging from 5 to 95 percent, the operating exposure time shall be a minimum of two hours at each of the temperatures specified in the applicable specification sheets.

3.9.6 Relative humidity. The receiver shall operate within all performance requirements under ambient atmospheric pressure (28.5 plus 2.0 inches minus 4.5 inches of mercury) and the following conditions of relative humidity applied in the following sequence:

a. Exposure to at least 95% relative humidity for a temperature of 29°C ± 4.0°C for a period of two weeks.

b. Exposure to at least 95% relative humidity for temperatures below 29°C consisting of two temperature cycles from 29°C to -40°C (±2.0°C) with a cycle period of three hours.

c. Exposure to a dew point of 29°C ±2.0°C at air temperatures above 29°C consisting of two temperature cycles from 29°C to 71°C (±2.0°C) with a cycle period of three hours.

3.10 Electromagnetic interference suppression.

3.10.1 Interference response rejection (antenna terminal). Using method CS04 and CS08, MIL-STD-462, the spurious response with the receiver tuned to any operating frequency, shall be within the limits specified in MIL-STD-461 limits for CS04 and CS08, as modified by the applicable specification sheet. The applicable specification sheet as a minimum shall specify the following:

a. The reference level for CS08 and CS04.

1/ Value specified in applicable specification sheet.

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b. Limit "A" for CS04 and CS08.

c. Limit "B" for CS04 and CS08.

3.10.2 Interference rejection (power lead). The receiver shall perform as specified herein when using Test Method CS01, MIL-STD-462, and when subjected to test levels specified in CS01, MIL-STD-461. Transients and interruptions in the input power leads shall not cause spurious output activation. Voltage transient levels and durations shall be as specified in the applicable specification sheet.

3.10.3 Conducted emissions. The conducted emissions shall be within the limits specified in MIL-STD-461 for CE01, CE02, CE03, CE04 and CE06 when using the respective test methods outlined in MIL-STD-462.

* 3.10.4 Radiated Emissions. The radiated emissions shall be within the limits specified in Mil-STD-461 for RE02 when using the test method outlined in MIL-STD-462.

* 3.10.5 Radiated Susceptibility. The receiver shall perform as specified herein when using Test Method RS03.2, MIL-STD-462, and when subjected to test levels specified in RS03.2, MIL-STD-461.

3.11 Physical characteristics.

3.11.1 Case. The command receiver shall be housed in an aluminum container, complete with mounted connectors, and have the dimensions and weight specified in the applicable specification sheets.

3.12 Identification. Each unit furnished under this specification shall be legibly marked and provide the following identification information:

a. Nomenclature.

b. Type, Class and Style Designation.

c. Manufacturer.

d. Unit Serial Number.

3.13 Workmanship. Workmanship shall comply with Requirement 9 of MIL-STD-454.

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3.14 Technical data. Technical data shall be furnished as set forth on the Contract Data Requirements List (CDRL), DD Form 1423, and in accordance with the data specification(s) referenced therein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification.

4.2.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification Inspection (see 4.4).

b. First Article Inspection (see 4.5).

c. Quality Conformance Inspection (see 4.6).

4.3 Inspection conditions. Unless otherwise specified for a specific test or series of tests, inspections shall be performed in accordance with the following test conditions:

a. No adjustment(s) of a receiver under test will be made during conduct of the tests.

b. Receiver input power: Specified nominal voltage $\pm$.5.

c. Ambient temperature: 23°C $\pm$10°C.

d. Ambient relative humidity: 5% to 70%.

e. Frequency tolerance, RF source: $\pm$2 kHz.

f. Deviation tolerance, RF source: $\pm$10%

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g. Tolerance of DC voltage range limits: ±0.2 volt.

h. Frequency tolerance, tone channels: ±0.1%.

i. Measurement resolution: Established by test equipment used (see 4.8.1).

j. Test equipment to be used is listed in Table IV.

4.4 Qualification inspection. Qualification inspection will be performed by the Government (see 6.5) on sample units submitted. These units shall be produced with equipment and procedures normally used in production.

4.4.1 Sample size. Two receivers shall be submitted for qualification inspection.

4.4.2 Inspection routine. Qualification inspection will be as specified in Table II, in the order shown.

4.4.3 Retention of qualification. Each manufacturer having a unit listed on the Qualified Products List (QPL) shall furnish certification for retention of the QPL. Certification shall be furnished at least every two years and shall be signed by a responsible official of management, attesting that the listed product(s) is (are) still available from the listed plant, can be produced under the same conditions as originally qualified; i.e., same process, material, construction, design manufacturer's part number or designation, and meets the requirements of the current issue of the specification. Failure to provide the certification will be cause for removal from the QPL. After completion of the certification review by the Government, the QPL will be reprinted to show the date of validation. (DD Form 1718, Certification of Qualified Products, shall be used for obtaining certification.)

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TABLE II. Qualification and first article inspection.

Inspection	Requirement paragraph	Test or examination method paragraph	
Physical inspection:			
Case	3.11.1	4.7.1	
Weight	3.11.1	4.7.2	
Case finish, construction workmanship	3.7.1, 3.13	4.7.3	
Electrical interface	3.7.2	4.7.4	
Identification	3.12	4.7.5	
Performance inspection:			
Input power	3.8.11.1, 3.8.11.2	4.8.1.1	
Threshold sensitivity	3.8.2, 3.8.4	4.8.1.2	
Output function	3.8.9	4.8.1.3	
RF operating level	3.8.5	4.8.1.4	
Receiver operation at power voltage limits	3.8.11.1	4.8.1.5	
RF level monitor	3.5.1.3	4.8.1.6	
Antenna impedance	3.8.6	4.8.1.7	
RF bandwidth, 3 db & 60 db	3.8.1	4.8.1.8, 4.8.1.9	
Quieting sensitivity, 20 db	3.8.3	4.8.1.10	
Audio monitor characteristics	3.5.1.4	4.8.1.11	
Decoder frequency char	3.8.7	4.8.1.12	
Decoder signal plus noise	3.8.8	4.8.1.13	
Channel response time	3.8.10	4.8.1.14	
Power control circuit	3.8.11.4	4.8.1.15	
Interference rejection, CS08	3.10.1	4.8.1.16	
Interference rejection, CS04	3.10.1	4.8.1.17	
Spurious response to power switching	3.10.2	4.8.1.18	
Spurious emission, RF input	3.10.3	4.8.1.19	
Spurious emission, power leads	3.10.3	4.8.1.20	
Spurious emission, control and signal leads	3.10.3	4.8.1.21	
Radiated Emission	3.10.4	4.8.1.22	*
Radiated Susceptibility	3.10.5	4.8.1.23	*
Environmental inspection:			
Temperature	3.9.5	4.8.2	
Vibration	3.9.4	4.8.3	
Shock	3.9.3	4.8.4	
Acceleration	3.9.1	4.8.5	
Altitude	3.9.2	4.8.6	
Humidity	3.9.6	4.8.7	
Performance stability	3.9	4.8.8	
Reliability	3.6	4.9	

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4.5 First article inspection. First article inspection shall be performed by the Government. First article inspection will be performed on sample units which have been produced with equipment and procedures to be used in production. First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts.

4.5.1 Sample size. The contract will set forth the number of units which are to be tested for first article inspection.

4.5.2 Inspection routine. First article inspection shall be as specified in Table II, in the order shown.

4.5.3 Data required. All first article testing performed shall be documented in the first article test report (see 6.2).

4.6 Quality conformance inspection. Quality conformance inspection shall be as set forth in Table III in the sequence shown.

4.7 Physical inspection.

4.7.1 Case. Measure unit dimensions and check configuration for conformance to the applicable specification sheet.

4.7.2 Weight. Weigh units to be tested. Unit weight shall meet the requirements of the applicable specification sheet.

4.7.3 Case finish, receiver construction and workmanship. Examine units for finish defects, dents, cracks or flaws.

4.7.4 Electrical interface. Check electrical interface connectors and their wiring for conformance to the applicable specification sheet.

4.7.5 Identification. Check identification tag or plate for required information and construction.

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TABLE III. Quality conformance inspection.

Inspection	Requirement paragraph	Test or examination method paragraph
Physical inspection:		
Case	3.11.1	4.7.1
Weight	3.11.1	4.7.2
Case finish, construction workmanship	3.7.1, 3.13	4.7.3
Electrical interface	3.7.2	4.7.4
Identification	3.12	4.7.5
Performance inspection:		
Input power	3.8.11.1, 3.8.11.2	4.8.1.1
Threshold sensitivity	3.8.2, 3.8.4	4.8.1.2
Output function	3.5.3, 3.8.9	4.8.1.3
Vibration environment	3.9.4	4.8.3
Temperature environment	3.9.5	4.8.2

4.8 Test methods. Performance tests are to be conducted to determine compliant operation in the various environments as required. The conditions of 4.3 are applicable to the performance tests unless specified otherwise.

4.8.1 Basic performance tests. Basic performance tests are to be performed in accordance with the following paragraphs. The test equipment listed in Table IV and the test configurations shown in Figures 2, 3, 4 and 5 are to be used for this testing.

4.8.1.1 Input power. Adjust the input power voltage to the nominal level and to the level at each end of the range specified in the applicable specification sheet. Measure and record the input current at each level for:

a. All command channels deactivated (tone off).

b. Each command function for which an input current level is specified. Requirement: Current levels to be within the limits specified in the applicable specification sheet.

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4.8.1.2 Threshold sensitivity. Set RF generator at minimum output, f_c frequency modulated $\pm$30 kHz per frequency by each of the command channel frequencies specified in applicable specification sheet. Increase RF level until the receiver functions in accordance with the applicable specification sheet. Change RF frequency $\pm$.005% and repeat test for f_c +.005% and f_c -.005%. Record RF power levels. Requirement: $\leq$ Level specified in applicable specification sheet.

4.8.1.3 Output functions. Connect RF generator to receiver RF input. Set RF generator at the measured threshold level (4.8.1.2) plus 3 db. Activate the command channels to provide the input functions specified in the applicable specification sheet. Monitor and measure the output functions required by the applicable specification sheet. Requirement: Functions in accordance with applicable specification sheet.

4.8.1.4 RF operating level. Set RF generator frequency to receiver operating frequency, f_c frequency modulated $\pm$30 kHz per tone, by each of the command channel tones specified in the applicable specification sheet and apply to receiver RF input. Adjust generator for the maximum receiver RF input level value specified in the applicable specification sheet. Decrease RF level to minimum receiver RF input value specified in the applicable specification sheet. Monitor the operation of the receiver for each of the above conditions. Requirement: Receiver shall function within performance requirements throughout the RF input level range specified in the applicable specification sheet.

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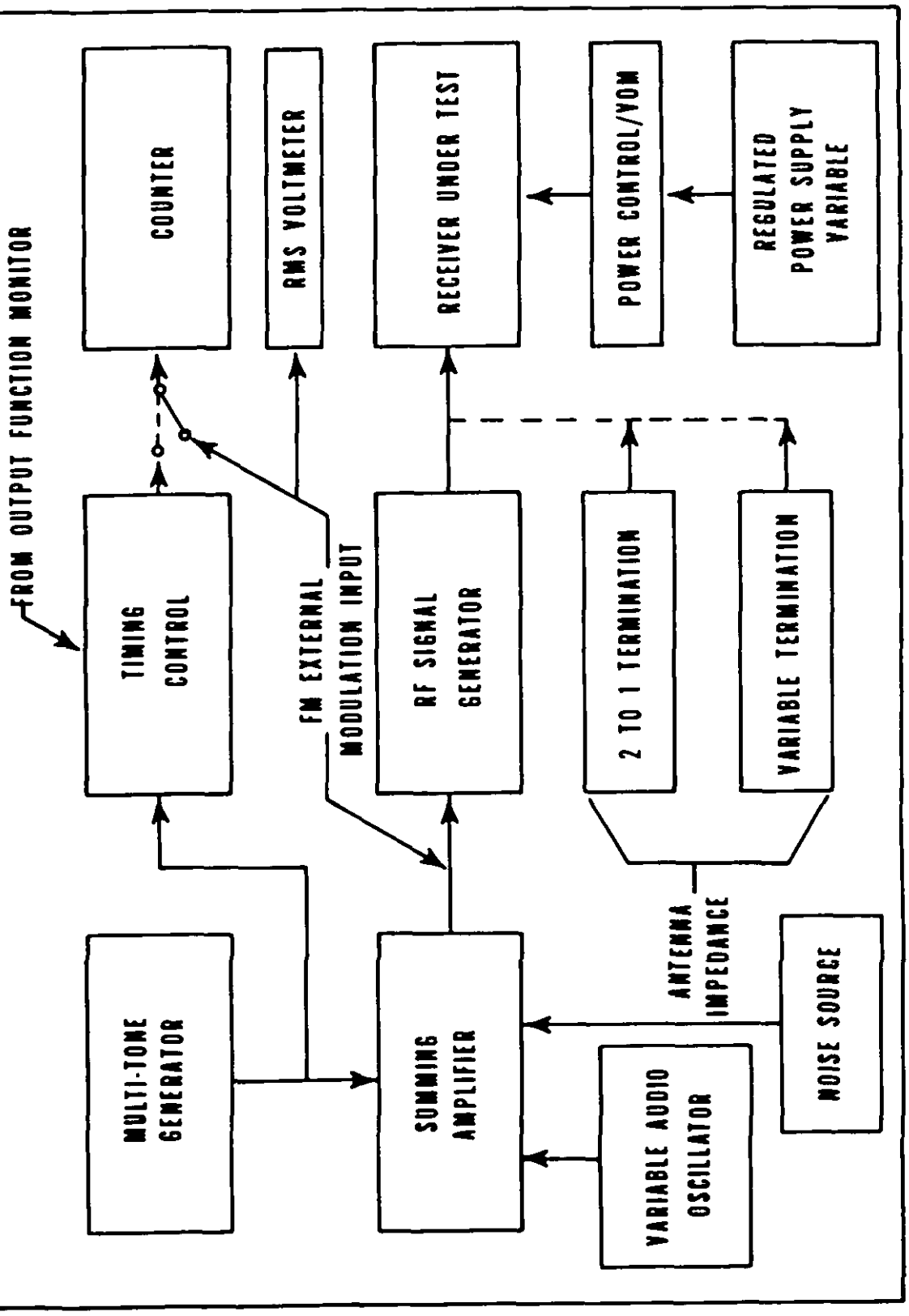


FIGURE 2 GENERAL TEST SETUP(INPUT CONFIGURATION)

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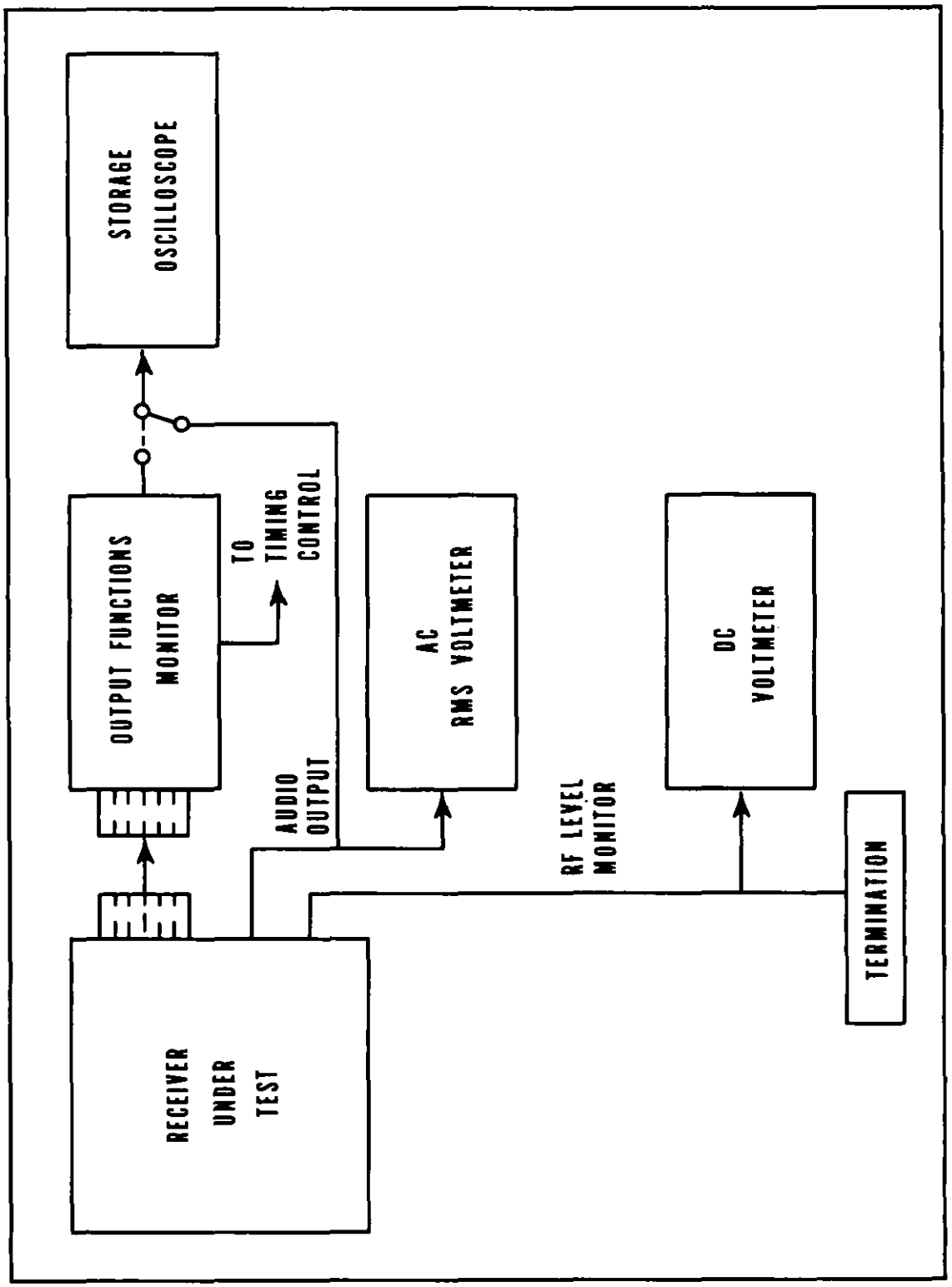


FIGURE 3. GENERAL TEST SETUP (OUTPUT CONFIGURATION)

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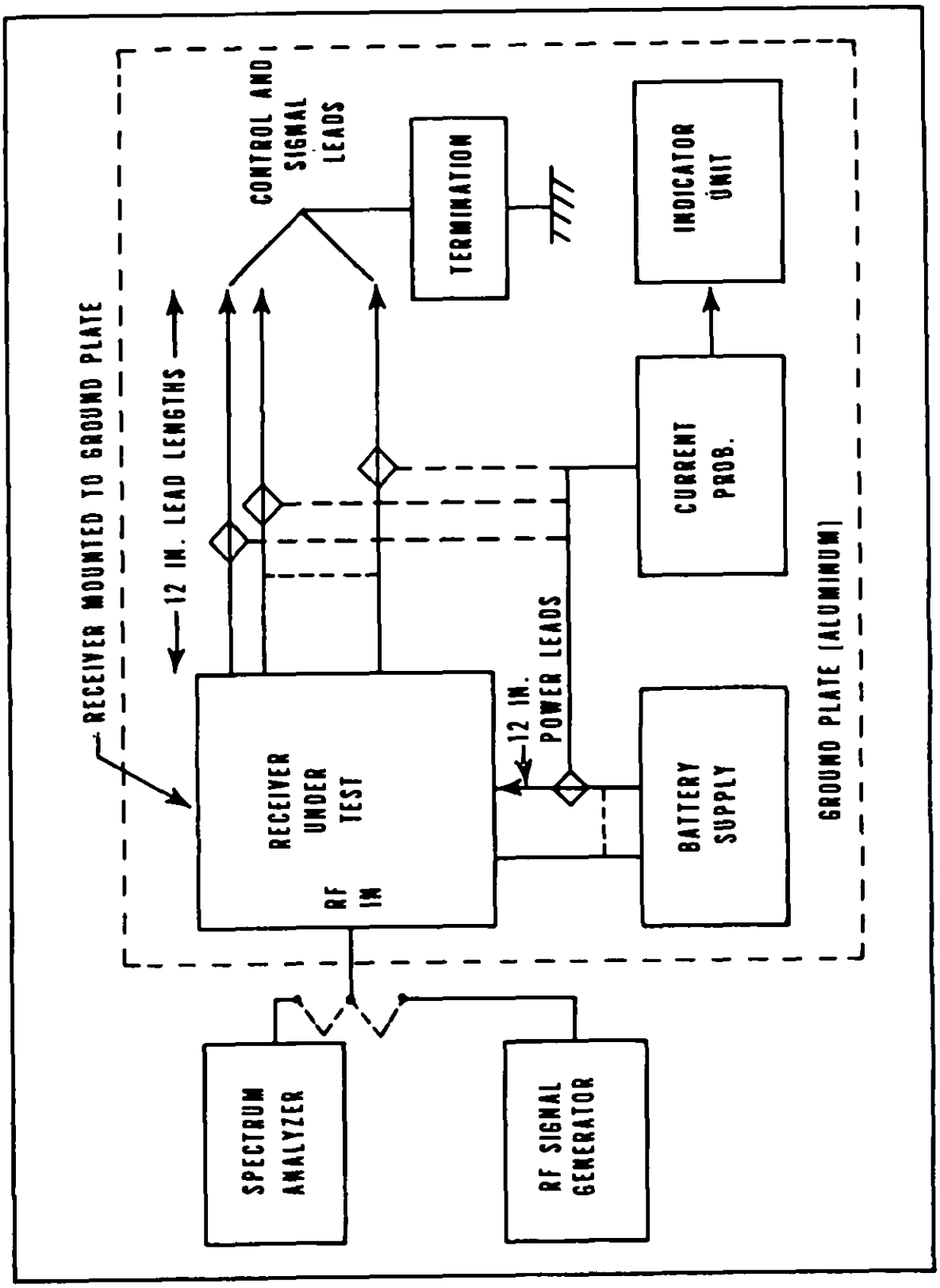


FIGURE 4. TEST SETUP FOR CONDUCTED EMISSIONS.

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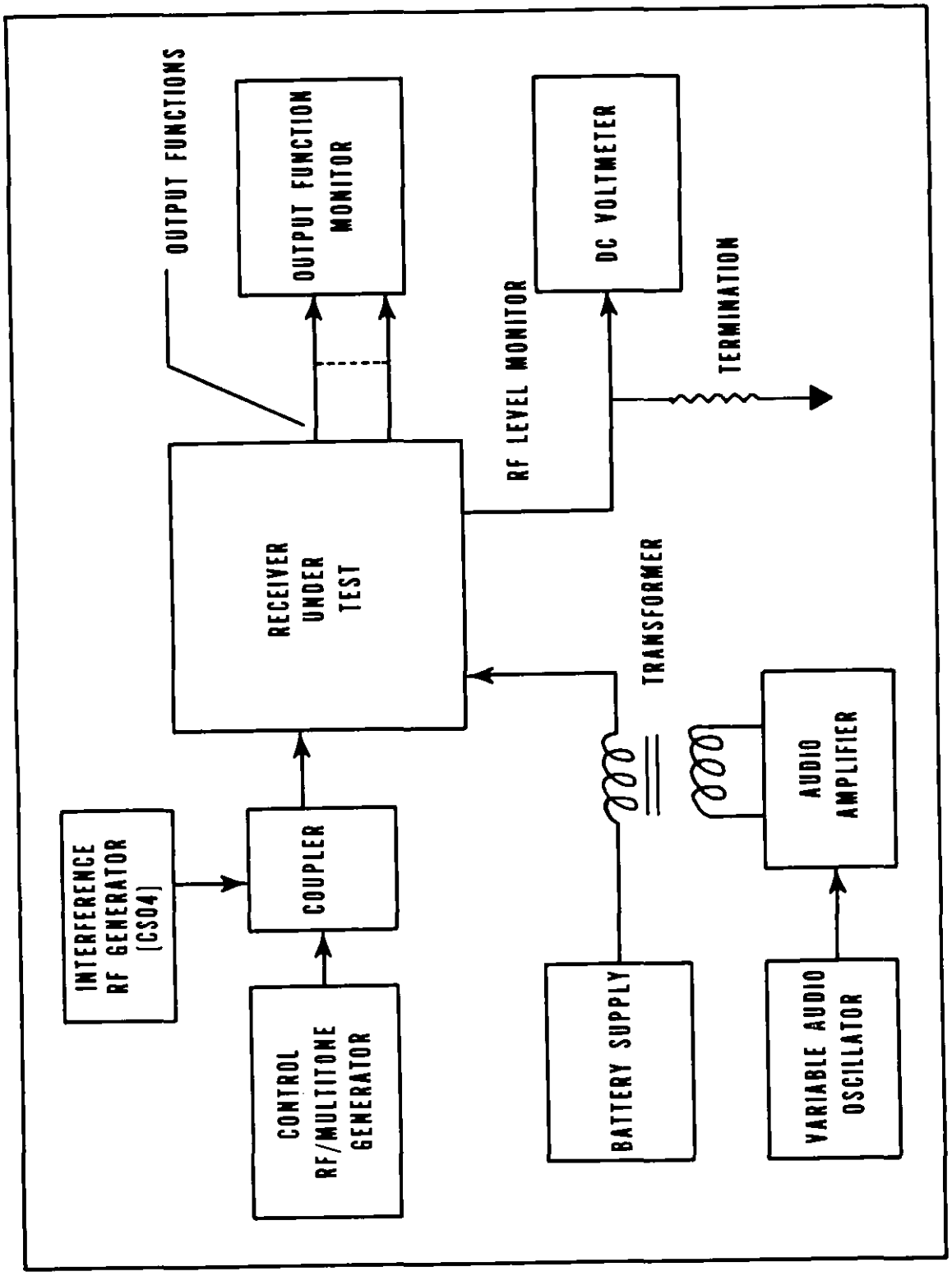


FIGURE 5. TEST SETUP FOR CONDUCTED SUSCEPTIBILITY.

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TABLE IV. Test equipment.

Test equipment (Figures 2 and 3) 1/		Model
RF signal generator	Hewlett-Packard	8660B
Multitone generator	Collins	AN/FRW 2
Variable audio oscillator	Hewlett-Packard	209A
RF termination, 2:1	Texscan	
Adjustable line	General Radio	874LA
Counter	Hewlett-Packard	5245L
Storage oscilloscope	Tektronix	564B
AC RMS voltmeter	Hewlett-Packard	3400A
DC voltmeter	Hewlett-Packard	5265A/5245L
VOM (digital)	Sustron-Donner	7050
Regulated power supply	Hewlett-Packard	6226B
Output function monitor		
Summing operational amplifier		741IC
Monitor circuitry w/power		
Test equipment (Figure 4, conducted emission) 1/		
Spectrum analyzer or receiver(s) (as per MIL-STD-462)		
Current probe (as per MIL-STD-462)		
Indicator unit (as per MIL-STD-462)		
Battery nickel cadmium (minimum capacity of 180 milliampere hour)		
VOM (digital) (same as Figure 2)		
Test equipment (Figure 5, conducted susceptibility) 1/		
Interference generator(s) (as per MIL-STD-462)		
Control RF and multitone generator (same as Figure 2)		
Battery nickel cadmium (minimum capacity of 180 milliampere hour)		
Variable audio oscillator (same as Figure 2)		
Coupler (as per MIL-STD-462)		
Audio amplifier and transformer (as per MIL-STD-462)		

1/ Test equipment may be substituted if it is determined by the Government to possess equal or improved essential capability and resolution.

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4.8.1.5 Receiver operation at input power voltage limits. For power supply voltages of the upper and lower limit of the input power voltage range specified in the applicable specification sheet repeat the threshold sensitivity test, 4.8.1.2, and the output functions test, 4.8.1.3. Requirement: Same as requirements for threshold sensitivity and output functions tests.

4.8.1.6 RF level monitor. Connect external load resistor, of the value specified in the applicable specification sheet, to the RF level monitor output. Use a voltmeter to read the DC output voltage across the resistor. Set the RF generator at f_c, CW, and apply to receiver RF input levels as follows:

-110 dbm
-105 dbm
-100 dbm
- 95 dbm
- 90 dbm
- 85 dbm
- 80 dbm and

values greater than -80 dbm up to maximum generator RF output level (approximately +13 dbm).

4.8.1.6.1 Requirements.

a. Maximum output level - 5 volts DC.

b. Linearity - as required in applicable specification sheet.

c. Repeatability - as required in applicable specification sheet.

d. Other - all other RF level monitor characteristics required by the applicable specification sheets.

4.8.1.7 Antenna impedance. With 50 ohm termination at receiver RF input, monitor the audio output, the RF level monitor output, and the output functions. While changing, in sequence, to no termination, a 2:1 termination, and while varying the length of the adjustable line termination, look for effects on the outputs. Record any response. Requirement: Unit shall not become unstable, oscillate or be damaged.

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4.8.1.8 RF bandwidth, 3 db. Set RF generator output power to reference level (see applicable specification sheet) at the receiver operating frequency, f_c, CW. Note reference on receiver RF level monitor. Increase generator output to upper and lower frequencies which produce this reference level and record each frequency. Requirement: BW_3 $\geq$ value required by applicable specification sheet; BW_3 center frequency within f_c, ± 0.2 BW_3.

4.8.1.9 RF bandwidth, 60 db. Set RF generator output power to reference level (see applicable specification sheet) at the receiver specified operating frequency f_c, CW. Note reference on RF level monitor. Increase generator output 60 db and detune generator to upper and lower frequencies that give this reference output on RF monitor. Count and record each frequency. Requirement: BW_{60} $\leq$ value required by applicable specification sheet; BW_{60} center frequency within f_c ± 0.2 BW_{60}.

4.8.1.10 Quieting sensitivity, 20 db. Set RF generator at minimum output, f_c, CW. Note audio output noise level. Increase RF carrier until noise level drops 20 db. Record RF level. Requirement: = -92 dbm.

Note: Cannot be performed with unit closed if audio monitor output is not required by applicable specification sheet.

4.8.1.11 Audio monitor output level and frequency response. Connect RF signal generator to receiver RF input. Set RF generator output power to level specified in applicable specification sheet, f_c, frequency modulated $\pm$ 30 kHz peak deviation with a variable signal frequency source. Monitor the receiver audio output level with an RMS voltmeter. Increase the modulating frequency from 7.5 kHz through the range specified in the applicable specification sheet. Record minimum and maximum output levels. Requirement: Set forth in applicable specification sheet.

4.8.1.12 Decoder frequency characteristics. Connect RF generator to the receiver RF input. Set RF generator at -95 dbm level, f_c, frequency modulated $\pm$15 kHz by a variable audio frequency source. Monitor output section functions. Vary the modulating frequency and measure the frequency band for each specified channel over which a "channel activated" decision is made by the decoder section. Record the upper and lower frequency of the measured frequency band. (This is the "response band" for that decoder channel.) Requirement: Each channel response band must cover the frequency range specified in the applicable specification sheet.

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4.8.1.13 Decoder signal plus noise. Set RF generator at f_c and -70 dbm output level and connect to receiver RF input. Separately activate command channel frequency and the noise source and adjust the deviation level of each such that:

a. Channel frequency level is one db below the "channel activated" decision level. Note the RMS value.

b. RMS noise level to 12 db above (1) above. Sum the signal and noise at the levels set in (1) and (2) and apply as modulation to the RF generator. Monitor the channel status of each channel with the above modulation conditions for 120 ±10 seconds. Requirement: "Channel activated" decision made less than 50% of the time.

b. Test II - Repeat the above for Test I except set channel frequency level at one db above the "channel activated" decision level. Requirement: "Channel activated" decision made more than 50% of the time.

4.8.1.14 Channel response time. Connect the RF generator to the receiver RF input and connect the audio signal modulating the RF generator to activate timer start. Connect the receiver output to operate timer stop. Measure the time between the turn-on of modulation and proper receiver output response. Measure each channel. Requirement: Specified in applicable specification sheet.

4.8.1.15 Power control circuit. Activate the power control circuit(s) with the specified control functions. Monitor and measure the output functions of the power control circuit(s) for operation as required by the applicable specification sheet. Requirement: Functions in accordance with requirements of the applicable specification sheet.

4.8.1.16 Interference rejection CS08 (antenna terminal). Using method CS08 of MIL-STD-462, apply RF test levels specified in the applicable specification sheet. All responses to the RF input test signal on the RF level monitor output of the receiver which exceed that caused by the specified reference level shall be recorded as a spurious response, provided the test signal frequency does not fall within the bandwidth of the receiver in accordance with MIL-STD-462. Requirement: No measured spurious responses permissable.

4.8.1.17 Interference rejection, CS04 (antenna terminal). Using method CS04 of MIL-STD-462, establish control of the receiver output functions by activating command channels with the control signal generator set at f_c ±.005% with an output level at the reference level specified for CS04, in the applicable specification sheet. Apply RF interference levels specified in the applicable specification sheet. Any loss of control by the control signal generator

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shall be recorded along with the corresponding frequency of the interference generator. Requirement: Interference levels to be within the limits specified in applicable specification sheet.

4.8.1.18 Spurious response to power input switching. Monitor all output functions and look for receiver operations as the DC power to the receiver is applied, removed, or interrupted. (Test with no RF signal applied.) Apply voltage transients as specified in applicable specification sheet to the receiver through the power lead and look for spurious output activation. Requirement: No measured activation of any output function permissable.

4.8.1.19 Spurious emission, conducted (RF input terminal). Measure conducted spurious emission with a receiver or spectrum analyzer having a nominal 50 ohm impedance, and calibrate each response against an RF generator. Signal strength measured across a 50 ohm termination connected at the receiver RF input shall not exceed limits specified in MIL-STD-461.

4.8.1.20 Spurious emission, conducted (power leads). Measure conducted emission on power input and output leads in the frequency range 30 Hz to 50 mHz per Methods CE01 and CE03 of MIL-STD-462. Emission limits are specified in MIL-STD-461.

4.8.1.21 Spurious emission, conducted (control and signal leads). Measure conducted emission on all leads not tested in the frequency range 30 Hz to 50 mHz per Methods CE02 and CE04 of MIL-STD-462. Emission limits are specified in MIL-STD-461. This measurement shall be made with the leads providing output functions inactive (not powered) where possible and connected to receiver ground.

* 4.8.1.22 Spurious Emission, Radiated. Measure radiated emission as per RE02, MIL-STD-462 with the receiver operating in the Figure 4 test configuration. Requirement: Levels to be within limits specified in MIL-STD-461.

* 4.8.1.23 Radiated Susceptibility. Using method RS03.2 of MIL-STD-462, establish control of the receiver output, RF level at threshold, paragraph 4.8.1.2, plus two db. Apply RF interference level specified in RS03.2, MIL-STD-461. Any loss of control by the control signal generator shall be recorded along with the corresponding frequency of the interference. Except for the RF input, the cable(s) to the receiver shall not be shielded and there shall be at least 12 inches of cable length before placing RF bypass or filtering. Requirement: Interference levels to be within MIL-STD-461.

4.8.2 Performance tests, temperature environment. Perform all tests listed in Table V in the temperature environment and in the sequence listed immediately after unit under test has been subjected to a ninety minute temperature soak at -40°C (±2°). Repeat this same test sequence after a ninety minute soak at 71°C (±2°). Repeat the same test after a ninety minute soak at 23°C (±10°).

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4.8.3 Performance tests, vibration environment. Subject each unit to the vibration environment specified in the applicable specification sheet for the class of inspection being performed. During vibration testing in each axis, monitor the performance as specified in 4.8.3.1 through 4.8.3.4. After vibration testing, conduct post vibration tests in 4.8.3.4.

4.8.3.1 Input power. Monitor the input current as in 4.8.1.1 with a VOM. Requirement: Current levels to be within the requirements specified in the applicable specification sheet.

4.8.3.2 Output functions. Perform test in 4.8.1.3 with the receiver subjected to the vibration environment and monitor output functions. Requirement: Must perform specified command functions without error.

4.8.3.3 RF level monitor. Monitor the RF level monitor output as in 4.8.1.6. Requirement: Level to be within the requirements specified in applicable specification sheet.

4.8.3.4 Post vibration tests.

a. PV-1 RF bandwidth, 3 db: Repeat 4.8.1.8

b. PV-2 threshold sensitivity: Repeat 4.8.1.2

c. PV-3 output functions test: Repeat 4.8.1.3

TABLE V. Temperature environment testing.

Test Number	Performance parameter	Test method
T-1	Input power	4.8.1.1
T-2	Threshold sensitivity	4.8.1.2
T-3	Output function test	4.8.1.3
T-4	RF operating level	4.8.1.4
T-5	Operation at input power voltage limits	4.8.1.5
T-6	RF level monitor	4.8.1.6, 4.8.1.6.1
T-7	Antenna impedance	4.8.1.7
T-8	RF bandwidth, 3 db	4.8.1.8
T-9	RF bandwidth, 60 db	4.8.1.9
T-10	Quieting sensitivity, 20 db	4.8.1.10
T-11	Audio monitor output level	4.8.1.11
T-12	Command channel activation	4.8.1.12
T-13	Power control circuit	4.8.1.14
T-14	Power control circuit	4.8.1.15
T-15	Interference rejection (CS08) except limit frequency range of test to 200 mHz to 600 mHz	4.8.1.16
T-16	Spurious response to power input switching except limit to power ON-OFF portion of test	4.8.1.18

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4.8.4 Performance test, shock environment. Subject each unit to the shock profiles specified in applicable specification sheet. During each shock exposure, perform 4.8.4.1 and after each exposure perform 4.8.4.2 and 4.8.4.3. After shock tests have been completed. perform test in 4.8.4.4.

4.8.4.1 Output function. Monitor output functions as in 4.8.1.3 with no command channels activated. Requirement: Output functions shall remain in the no command channels activated condition.

4.8.4.2 Input power. Repeat 4.1.1.

4.8.4.3 Output function. Repeat 4.1.3.

4.8.4.4 Post shock test.

PS-1	Threshold sensitivity:	Repeat 4.1.2
PS-2	RF level monitor:	Repeat 4.1.6

4.8.5 Performance tests, acceleration environment. Subject each unit to acceleration levels specified in applicable specification sheet. During testing in each axis, monitor the performance as specified in 4.8.5.1 and 4.8.5.2. After acceleration testing perform 4.8.5.3.

4.8.5.1 Input power. Monitor the input current as in 4.8.1.1 with a VOM. Requirement: Current levels to be within the requirements specified in applicable specification sheet.

4.8.5.2 Output functions. Perform 4.8.1.3 and monitor output functions except establish RF input through an antenna coupled RF link and set the transmitter level at 6 db above the level required to establish threshold operation specified in 4.8.1.2. Requirement: Must perform specified command functions without error.

4.8.5.3 Post acceleration test.

PA-1	Threshold sensitivity:	Repeat 4.8.1.2
PA-2	RF level monitor:	Repeat 4.8.1.6

4.8.6 Performance test, altitude environment. Subject each unit to the altitude environment specified in applicable specification sheet. While subjected to the minimum pressure specified, conduct the following tests:

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a. Input power: Repeat 4.8.1.1

b. Threshold sensitivity: Repeat 4.8.1.2

c. Output functions: Repeat 4.8.1.3

4.8.7 Performance tests, relative humidity environment. Expose each unit to each of the relative humidity-temperature environments. (See 3.9.6.) Near the end of each exposure prior to completion of the exposure conduct the following:

a. Input power: Repeat 4.8.1.1

b. Threshold sensitivity: Repeat 4.8.1.2

c. Output functions: Repeat 4.8.1.3

d. RF level monitor: Repeat 4.8.1.6

4.8.8 Performance stability test. After all environmental performance tests in 4.8.2 through 4.8.7 have been completed, repeat 4.8.2 at ambient conditions as defined in 4.3 c and d.

4.9 Reliability demonstration. Reliability is to be demonstrated by employing a sequential reliability test conducted in accordance with Figure 7 where it is shown that if no failures are detected during the first 69,325 operation cycles or if the t_a line is crossed, the test item has passed. On the other hand, if three failures occur prior to 34,625 operation cycles or if the t_r line is crossed, the test item has failed. The test is forced to completion after 496,500 operation cycles or when 14 failures have occurred. The operating characteristic curve, Figure 8, shows the probability of passing the test as a function of the true mean number of operation cycles between failure. The reliability test setup is shown in Figure 6.

4.9.1 Conditions. The following conditions apply to the reliability testing.

a. The reliability demonstration test is to be conducted as a separate test after all other required testing is completed.

b. The total number of operation cycles and the total number of failures is the sum of these respective factors for all receivers included in the test.

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c. An operation cycle consists of activating all tone channels (tone on) for 2±0.5 seconds and deactivating all tone channels (tone off) for 3±0.5 seconds.

d. Throughout the reliability test, operation of the receiver output section is to be monitored.

e. A failure is defined as improper operation of the output section during an operation cycle.

f. Replacement of failed components or adjustment of the receiver under test will be allowed only as corrective action following a failure. All time utilized for such corrective action is to be exempted from the receiver reliability test time.

g. The RF signal level at the receiver input is to be maintained at -95±5 dbm throughout the reliability test.

h. Each channel tone signal is to be adjusted to a level which causes a 30±3 kHz peak deviation of the RF signal.

5. PACKAGING. Preservation, packing and marking of shipments shall be as set forth in Federal Standard 356.

6. NOTES

6.1 Intended use. The receivers covered by this specification will be used to control or terminate the flight of unmanned vehicles to protect personnel and property.

6.2 Ordering data. Procurement documents referencing this specification should specify the following:

a. Title, number and date of this specification.

b. Applicable specification sheet.

c. Style.

d. Operating RF frequency.

e. Channel frequencies. Specify channel selected from Table I for each decoder channel required by the specification sheet.

f. Any special packing, packaging or marking requirements.

g. Data required in first article test report, if article inspection is assigned to contractor.

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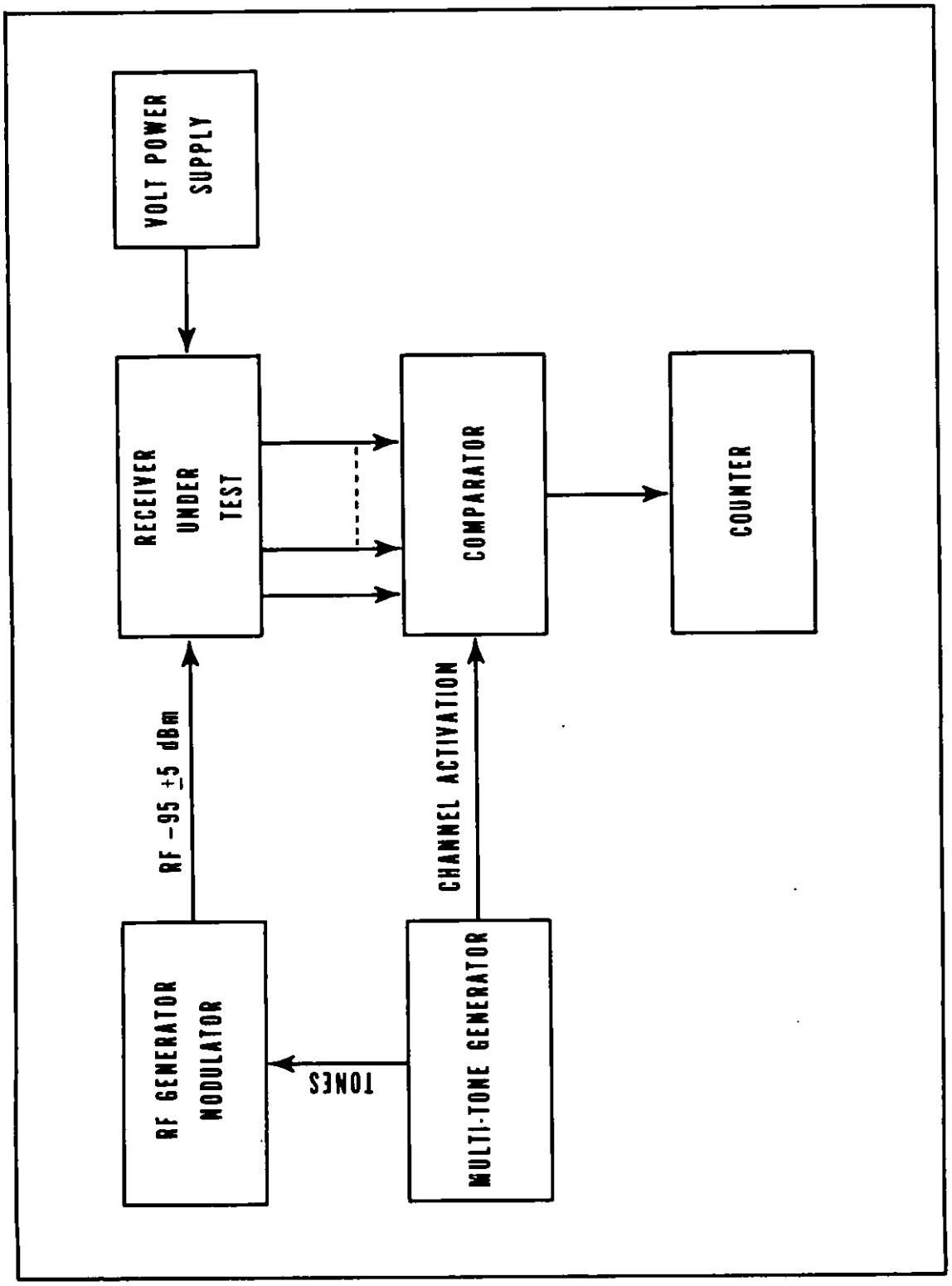


FIGURE 6. TEST SET-UP FOR RELIABILITY

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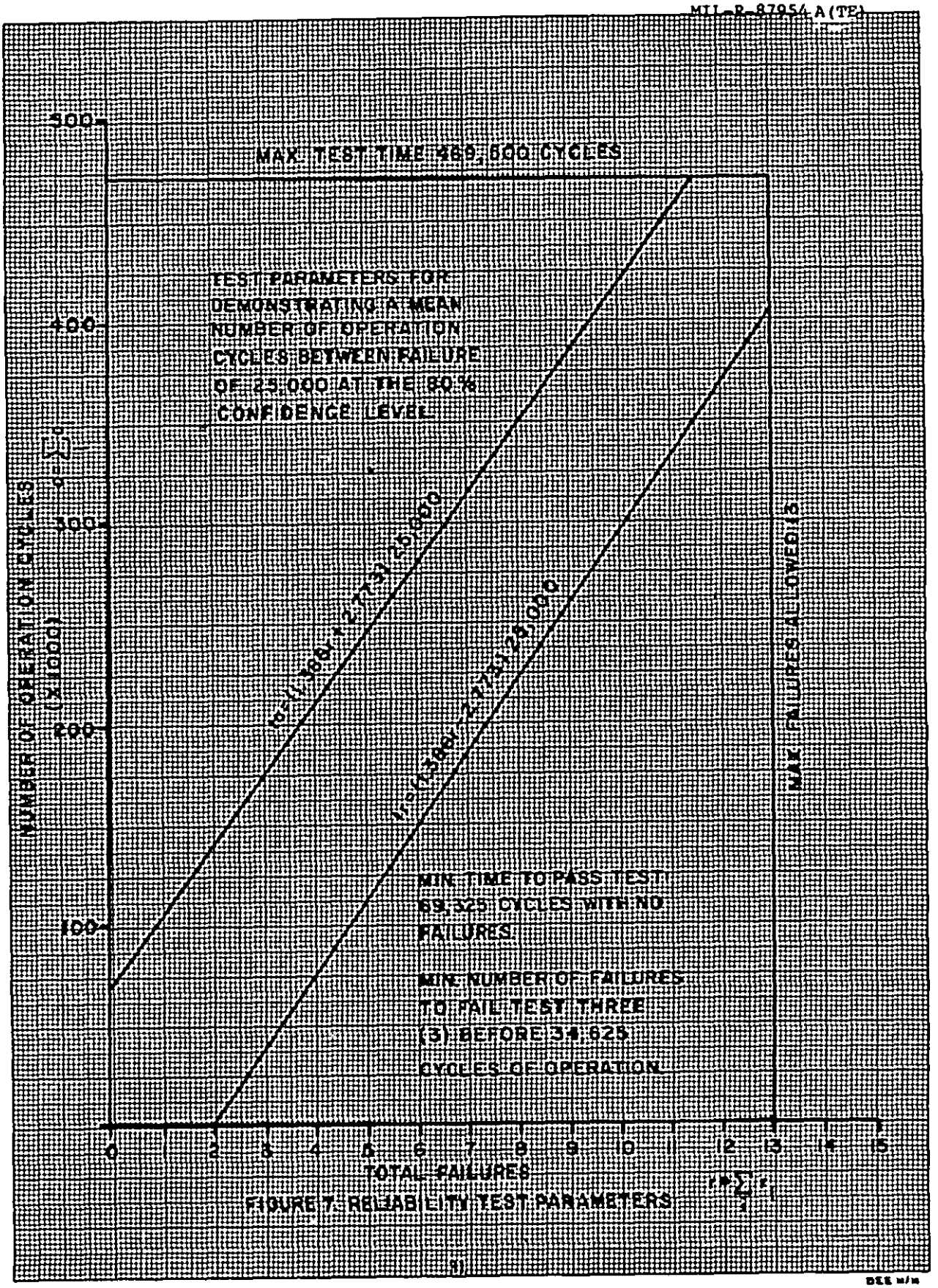


FIGURE 7. RELIABILITY TEST PARAMETERS

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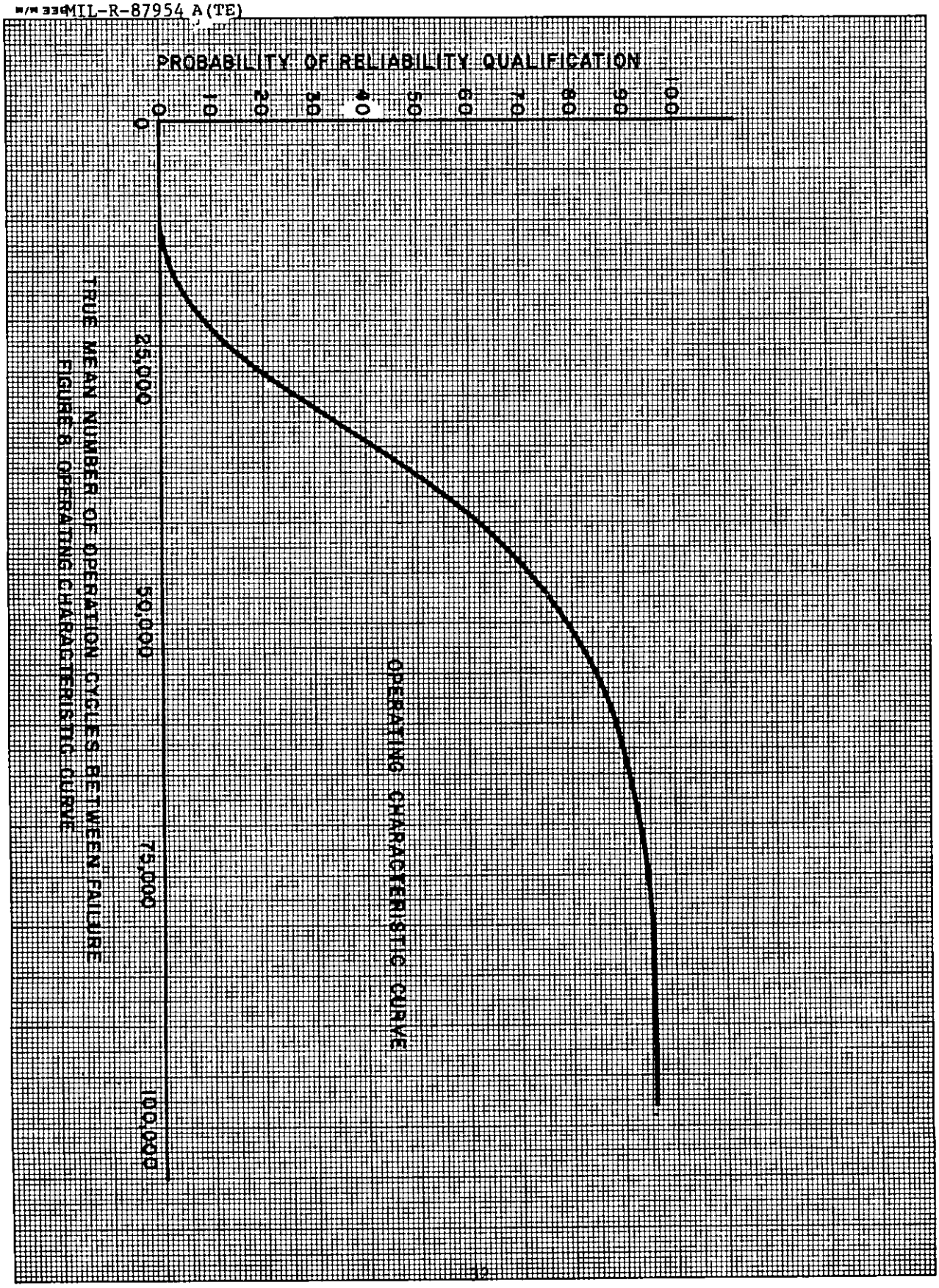


FIGURE 8. OPERATING CHARACTERISTIC CURVE.

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6.3 Technical data. When this specification is used in procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulation (DAR) the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification are cited in the following paragraph:

Paragraph	Data Requirement	Applicable DID
3.14	Operation and Maintenance Manual	DI-M-5104

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation (DAR). The first article shall be a first production item or it may be a standard production item from the contractor's current inventory.

6.5 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the US Army Test and Evaluation Command, DRSTE-AD-M, Aberdeen Proving Ground, Maryland 21005 and information pertaining to qualification of products can be obtained from that activity.

Preparing activity:
Army-TE

Project 1430-A119

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